

The Three Four, PA2

Members:

Sirjana Acharya, Summer Driver, Harper Laskie, Callie Sproles

GitHub URL

https://github.com/csproles/Project1_3400.git

Data Snippet

1	title	artist	top genre	year	bpm
2	Hey, Soul Sister	Train	neo mellow	2010	97
3	Love The Way You Lie	Eminem	detroit hip hop	2010	87
4	TiK ToK	Kesha	dance pop	2010	120
5	Bad Romance	Lady Gaga	dance pop	2010	119
6	Just the Way You Are	Bruno Mars	pop	2010	109
7	Baby	Justin Bieber	canadian pop	2010	65
8	Dynamite	Taio Cruz	dance pop	2010	120
9	Secrets	OneRepublic	dance pop	2010	148
10	Empire State of Mind (Part II) Broken Down	Alicia Keys	hip pop	2010	93
11	Only Girl (In The World)	Rihanna	barbadian pop	2010	126
12	Club Can't Handle Me (feat. David Guetta)	Flo Rida	dance pop	2010	128
13	Marry You	Bruno Mars	pop	2010	145
14	Cooler Than Me - Single Mix	Mike Posner	dance pop	2010	130
15	Telephone	Lady Gaga	dance pop	2010	122
16	Like A G6	Far East Movement	dance pop	2010	125

Contributors



Participation/Workload Distribution:

Each member did either one parent or child class to distribute work evenly in a cooperative manner.

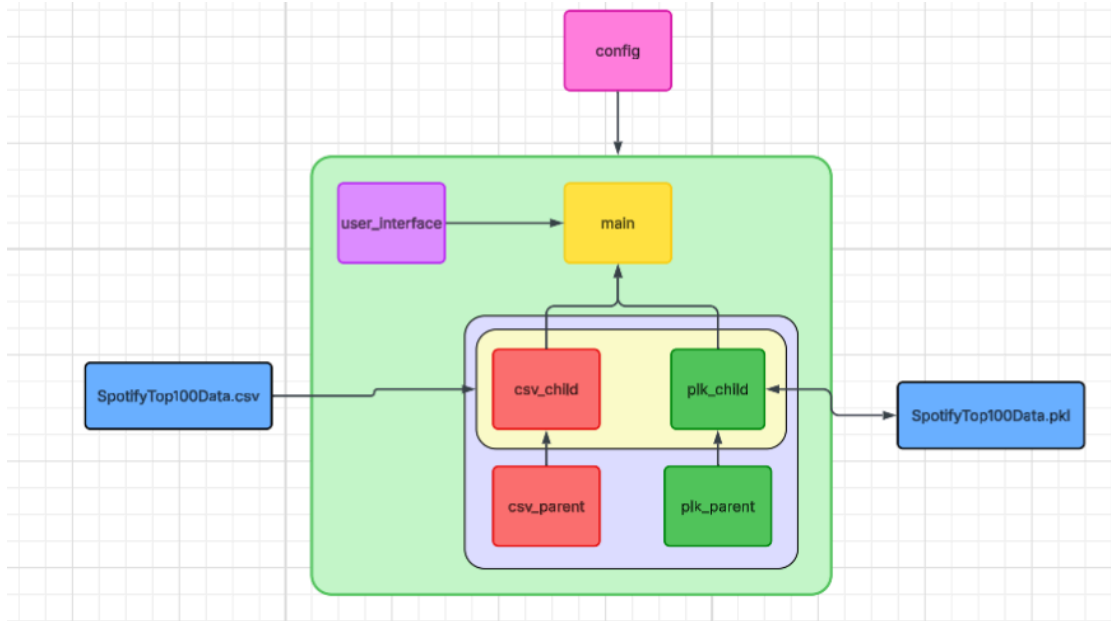
Harper - Parent 1

Summer - Child 1 (CSV)

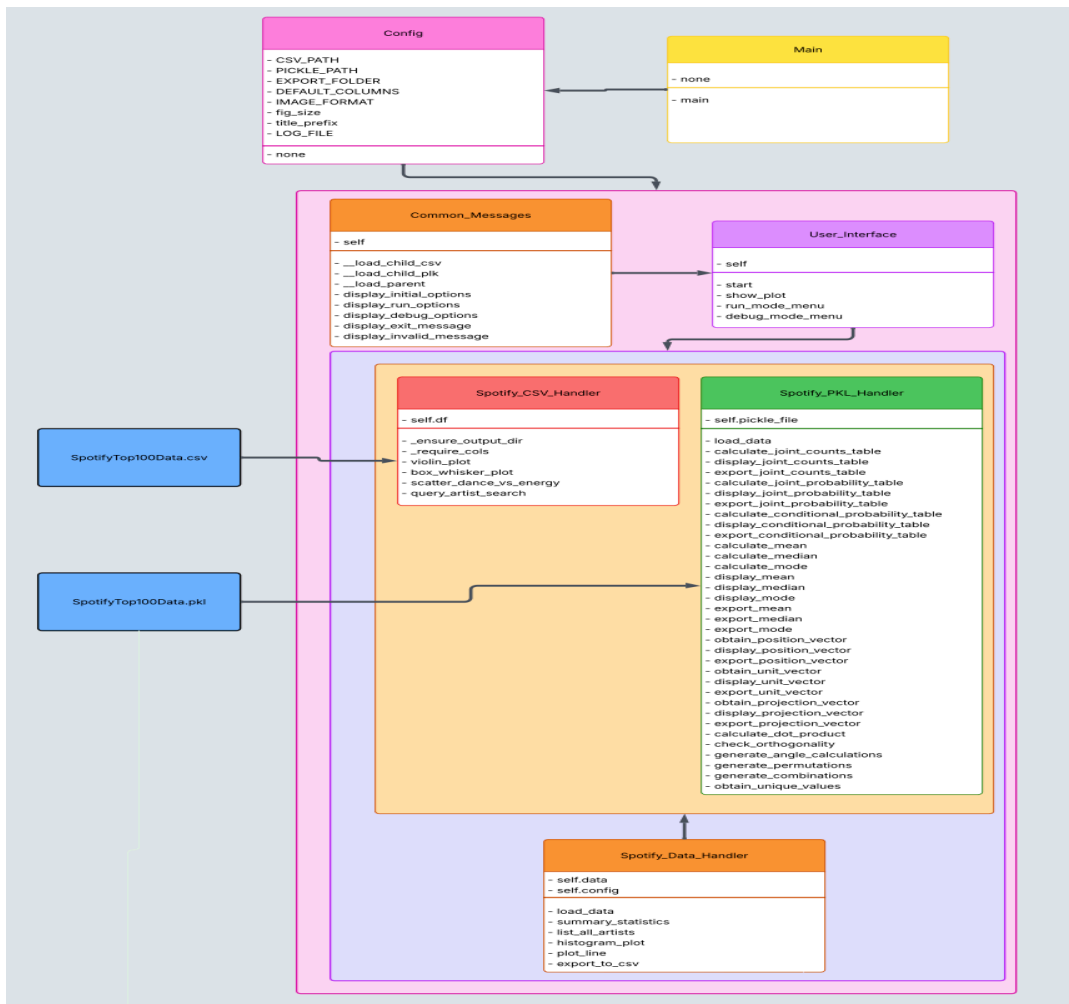
Sirjana - Parent 2

Callie - Child 2 (Pickle)

Module Flow Diagram Link



Class Diagram



Task Report:

11/4/2025	Created parent class 2 outline	Completed	Harper Laskie
11/4/2025	Created PA2 document	Completed	Harper Laskie
11/4/2025	Parent class one	Completed	Harper Laskie
11/4/2025	Completed draft expanded code for parent class 1	Completed	Harper Laskie
11/6/2025	Parent class one committed/pushed	Completed	Harper Laskie
11/6/2025	Added Summer to everything	Completed	Harper Laskie
11/7/2025	Child Pickle Class	Completed	Callie Sproles
11/7/2025	New data in main	Completed	Callie Sproles
11/7/2025	Created pickle data file	Completed	Sirjana Acharya
11/7/2025	Updated the configuration file as needed for now	Completed	Sirjana Acharya
11/7/2025	Moving to the discord groupchat	Completed	Sirjana Acharya
11/7/2025	Added .gitignore file to ignore unnecessary caches/files	Completed	Sirjana Acharya
11/7/2025	Parent Class 2 ready to be implemented by child pickle c	Completed	Sirjana Acharya
11/7/2025	Violin plot - child csv	Completed	Summer Driver
11/7/2025	Whisker plot - child csv	Completed	Summer Driver
11/7/2025	Scatter plot - child csv	Completed	Summer Driver
11/7/2025	Query w/Boolean indexing - child csv	Completed	Summer Driver
11/7/2025	Child Pickle - Joint Probability	Completed	Callie Sproles
11/8/2025	Child Pickle - Conditional Probability	Completed	Callie Sproles
11/9/2025	Child Pickle - Permutations	Completed	Callie Sproles
11/9/2025	Child Pickle - Combinations	Completed	Callie Sproles
11/9/2025	Child Pickle - Dot product	Completed	Callie Sproles
11/9/2025	Child Pickle - Projections	Completed	Callie Sproles
11/9/2025	Child Pickle - Angle Calculations	Completed	Callie Sproles
11/10/2025	Fixing overall code and visualization errors	Completed	Harper Laskie
11/10/2025	Readjusting Input for broader scope	Completed	Callie Sproles

Goal of the Project:

The goal of this project for parent 1 and child 1 is to load a Spotify dataset, convert it into a dataframe, and then analyze the data using multiple types of visualizations. By doing this, the program demonstrates data processing, plotting, and filtering techniques in python using object-oriented inheritance between a parent and child class. For parent 2 and child 2 their goal is to support more advanced mathematical and probability-based analysis by converting the dataset into a pickle file for faster access. Focusing more on the numeric operations they are allowing deeper statistical exploration beyond simple visual plots.

Code descriptions and output screenshots:

Main:

Code entry and execution. Creates instances of child classes, childCSV and PickleChild, to load our data, run queries, and perform calculations or visualizations. The main class also runs testing methods, like generating our projections and angles for PickleChild and creating plots and searching artists for childCSV. Includes a placeholder for a user interface, allowing input to be queried in our dataset.

Parent 1:

General class for working with CSV data. It serves to load a dataset into a pandas DataFrame, perform basic queries on columns, and generate visualizations like histograms and line plots. It also provides utility methods to list unique artists or genres and to display the top 10 artists by song count. Essentially, it handles common data loading, querying, and plotting tasks so child classes are able to extend its functions.

Child 1:

ChildCSV is a child class of ParentClass designed specifically for handling CSV data. It generates all required visualizations and boolean-based artist search. Child class CSV creates violin plots, box-and-whisker plots, and scatter plots with jitter and various data mapping. It also has the option to search for a specific artist to see if they are on the list so you do not have to scroll through and look for them. Automatically creating an output directory to save plot pngs and CSVs for your convenience.

Parent 2:

Provides tools for analyzing categorical and numerical data, and performs vector operations. Leaves data loading to child classes, allowing flexibility for specialized analysis.

Child 2:

PickleChild handles the pickled version of the dataset. It loads the CSV file into a pickle file, and provides methods for probability calculations, numerical vector analysis, and combination-based operations for mathematical tasks. Specifically, pickleChild:

1. Generates joint and conditional probabilities
2. Converts columns to arrays, dot products, projections, and angle calculations.
3. Gets unique values, counts of occurrences, and list conversions.
4. Generates permutations and combinations

Project Manual:

- ❖ The input data format for this project is a CSV file where each row represents a song and each column represents an attribute such as title, artist, genre, year, and BPM (beats per minute).
- ❖ The output data format includes plots, graphs, lists, projection and calculations.
- ❖ The user will get to choose a number between 1-12 from the user interface menu. Each number is associated with a different process directed by the program. They can continue picking numbers until they are satisfied with the outputs they have received. After they are finished they can pick 13 to exit the program.

Output (via Main):

Parent Class 1 Test output:

```
Top 10 Artists:
  artist
Lady Gaga      8
Kesha          6
Rihanna        6
Bruno Mars     5
The Black Eyed Peas  5
Beyoncé        4
Christina Aguilera  4
Britney Spears  4
Jennifer Lopez  4
OneRepublic    3
Name: count, dtype: int64

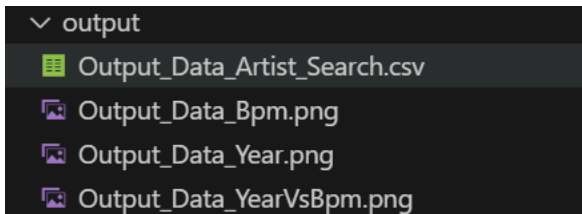
All processing complete.
```

```
Unique artists: ['Train' 'Eminem' 'Kesha' 'Lady Gaga' 'Bruno Mars' 'Justin Bieber'
'Taio Cruz' 'OneRepublic' 'Alicia Keys' 'Rihanna' 'Flo Rida'
'Mike Posner' 'Far East Movement' 'Usher' 'Sean Kingston'
'The Black Eyed Peas' 'Adam Lambert' 'Maroon 5' 'Neon Trees'
'Selena Gomez & The Scene' 'Enrique Iglesias' 'Katy Perry'
'Britney Spears' '3OH!3' 'David Guetta' 'Christina Aguilera'
'Florence + The Machine' 'Shakira' 'Tinie Tempah' 'T.I.' 'Martin Solveig'
'Christina Perri' 'Adele' 'Pitbull' 'Beyoncé' 'Hot Chelle Rae'
'Avril Lavigne' 'Kanye West' 'LMFAO' 'Jessie J' 'Jennifer Lopez'
'Chris Brown' 'Sleeping At Last' 'Nicki Minaj' 'P!nk']
```

```
Unique genres: ['neo mellow' 'detroit hip hop' 'dance pop' 'pop' 'canadian pop' 'hip pop'
'barbadian pop' 'atl hip hop' 'australian pop' 'indie pop' 'art pop'
'colombian pop' 'big room' 'british soul' 'chicago rap' 'acoustic pop']
```

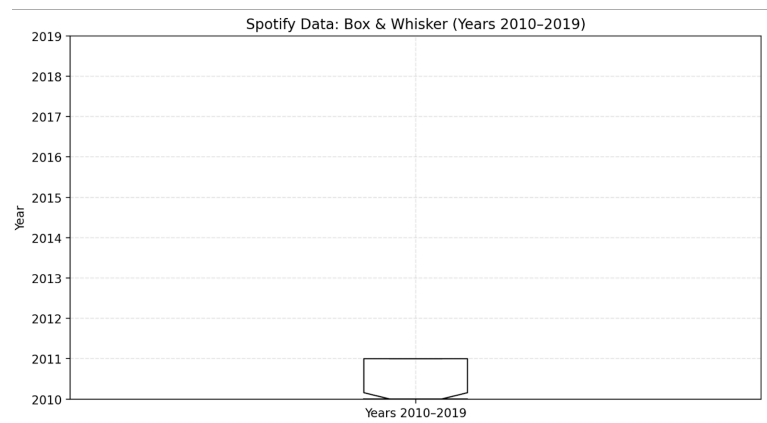
Child class 1 test output:

Output files:

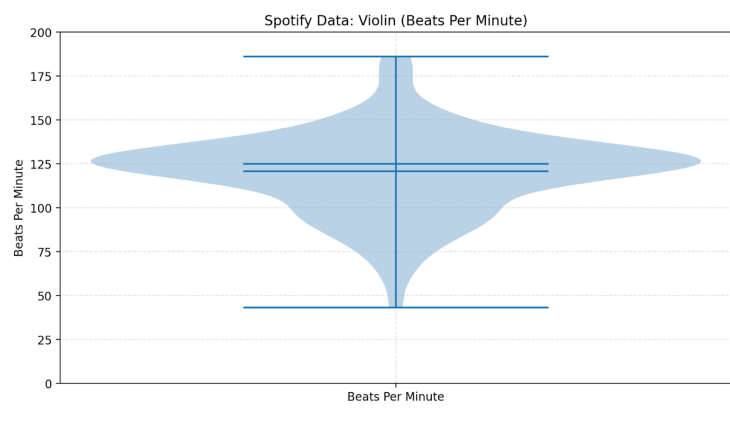


Plots:

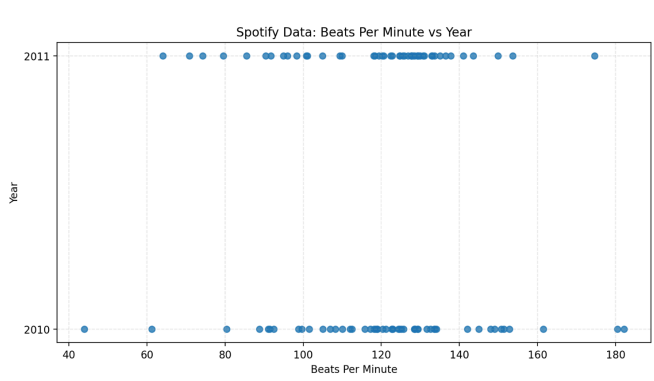
Box and Whisker



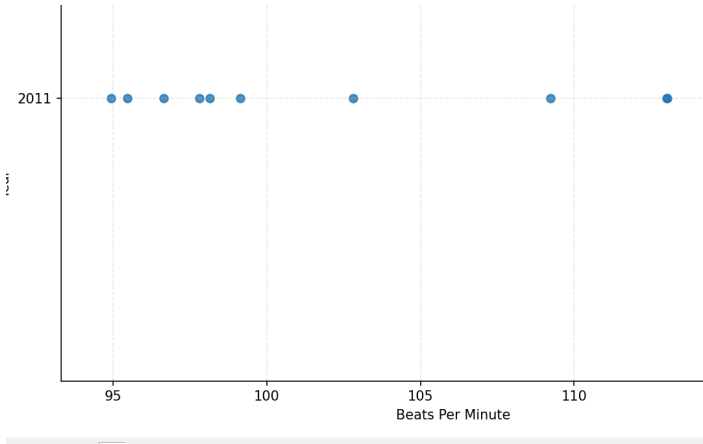
Violin:



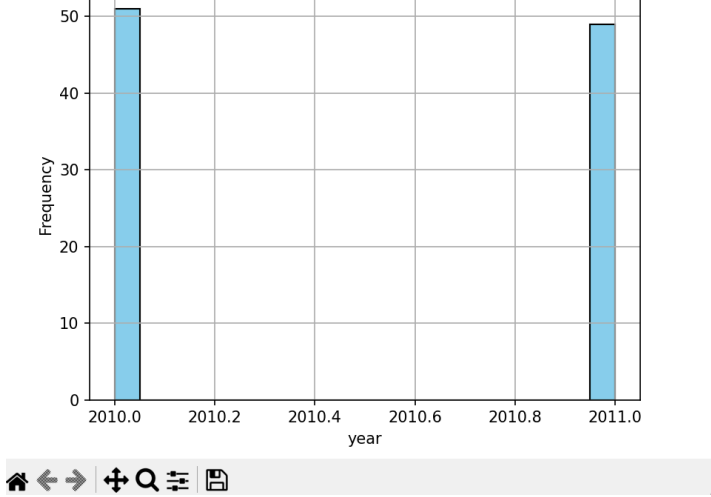
Scatter ----->



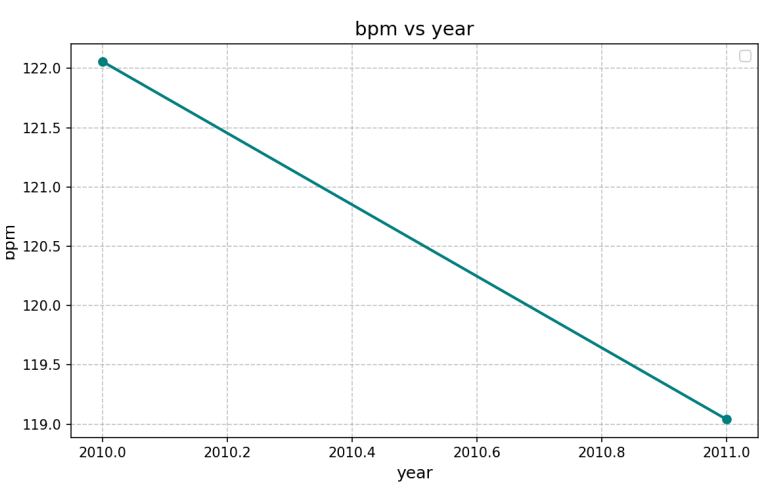
Zoomed scatter example:



Histogram of year



Smoothed Line Plot



Enter an artist name to search in childCSV: usher

Saved artist search CSV (2 rows) -> C:\Users\haroe\OneDrive\Documents\GitHub\Project1_3400\output\Output_Data_Artist_Search.csv

Parent/Child pickle output:

```
Dot Product: 24242413
Projection: [1558.96041265 1398.24284434 1928.61081978 1912.53906295 1751.82149463
1044.66419405 1928.61081978 2378.62001106 1494.67338533 2025.04136077
2057.18487443 2330.40474056 2089.32838809 1960.75433344 2008.96960393
2089.32838809 1944.68257661 2057.18487443 1591.10392632 1928.61081978
2089.32838809 2989.34677065 2008.96960393 1655.39095364 2346.4764974
2378.62001106 2137.54365859 2073.25663126 1928.61081978 2008.96960393
2169.68717225 2217.90244274 1928.61081978 1478.6016285 1783.96500829
2089.32838809 2635.76812036 2105.40014492 2008.96960393 2410.76352472
2410.76352472 1671.46271047 2925.05974333 1800.03676513 691.08554375
1928.61081978 1462.52987166 1928.61081978 1285.74054652 1960.75433344
2057.18487443 2233.97419957 2169.68717225 2073.25663126 1751.82149463
1687.5344673 2041.1131176 2057.18487443 1510.74514216 1767.89325146
1607.17568315 2410.76352472 1992.8978471 2008.96960393 2330.40474056
1591.10392632 2089.32838809 1928.61081978 2812.55744551 1526.81689899
1510.74514216 2089.32838809 1607.17568315 2089.32838809 2057.18487443
2089.32838809 2057.18487443 2442.90703838 2121.47190175 2089.32838809
1928.61081978 2041.1131176 2105.40014492 2025.04136077 1928.61081978
2057.18487443 2105.40014492 1012.52068038 2137.54365859 1350.02757384
1928.61081978 1285.74054652 2057.18487443 1462.52987166 1205.38176236
1285.74054652 2008.96960393 2041.1131176 1960.75433344 2073.25663126]
Angle Calculations: 86.5616085169049
```

Checklist

GROUP NAME	TASK NAME	MODULE NAME	LINE NUMBERS
General	Query DataFrame	ParentClass	111-141
	Write an mxn Numpy array into a dataframe	PickleChild	448-472
	Lambda function	user interface	39-59, 85-154
	Eval()		
	Logging	user interface	12-16
Function Parameters			
	*arg	user interface	32-33
	**kwarg	user interface	32-33
Variable Types	Default argument	pickle child	62,52
	Immutable	ParentClass	22-43
	Mutable	PickleChild	74-103
	Global	main()	1-7
Parent class-1	Nonlocal		
	Private like		
	Store configuration constants in a dictionary	ParentClass	45-65
	Histogram	plot_histogram()	150-205
Child class-1.1	Line plot	plot_line()	209-313
	Simple search	basic_query()	111-149
	Read data from a csv file. Store into a dataframe.	childCSV. __init__()	15-16
	Utilize configuration constants	childCSV	48
Child class-2.1	Violin plot	violin_plot()	38-63
	Whisker-box plot	box_whisker_year()	65-93
	Scatter plot	scatter_dance_vs_e	95-121
	Query w/Boolean indexing	query_artist_search()	123-153
	Read data from a pickle file	vs	94
	Utilize configuration constants	load_data()	85-88
	Visualize		
	Probability: Calculate, display, export		
	joint counts	calculate_joint_coun	97-145
	joint probabilities	calculate_joint_prob	147-196
	conditional probabilities	calculate_conditiona	198-250
	mean, median, std	calculate_mean(), calc	253-384
	Vector operators: Display, export		
	Obtain position vector	obtain_position_vect	400-412
	Obtain unit vectors	obtain_unit_vector()	423-437
	Obtain projection vectors	obtain_projection_ve	448-476
	Calculate the dot product	calculate_dot_produ	482-516
	The angle between two vectors	generate_angle_cal	540-572
	Check for orthogonality	check_orthogonality	518-534
	For a categorical attribute do the following and display		
	Obtain unique values	obtain_unique_value	626-637
	Generate permutations	generate_permutatio	574-599
	Generate combinations	generate_combinatio	601-624